Balance-Sheet Capacity and Cross-Strategy Treasury Arbitrage: SLR Exclusion Event Study

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Abstract

Regulatory leverage constraints are often described as shaping financial intermediaries' willingness and ability to intermediate U.S. Treasury markets, especially during stress episodes. The empirical focus of this study is the Federal Reserve's temporary exclusion of U.S. Treasury securities and Federal Reserve deposits from the Supplementary Leverage Ratio (SLR) denominator—effective April 1, 2020 and expiring March 31, 2021—as a time-dated, plausibly exogenous shock to balance-sheet capacity for covered bank holding companies. Using a pooled difference-in-differences design across 17 daily arbitrage spread series in four strategy classes (Treasury spot-futures, TIPS-Treasury, covered interest parity, and equity spot-futures), I test whether the SLR relief differentially compressed Treasury-based arbitrage spreads relative to non-Treasury controls.

The expiry of the exclusion is the paper's cleanest identification of price impact on arbitrage spread series. When the relief expired, Treasury-based spreads widened by 3.84 basis points more than non-Treasury spreads in the subsequent 60 trading days following SLR announcement. A full-period difference-in-differences using the 2019 pre-COVID baseline recovers -10.76 basis points of differential Treasury compression during the relief window. Non-Treasury series show no comparable break at the SLR event dates, consistent with balance-sheet segmentation rather than a general intermediary-health shock. These results provide direct policy-event evidence for the balance-sheet channel in Treasury arbitrage markets, complementing Siriwardane et al. [1], who document pervasive segmentation across 32 arbitrage strategies, and Bräuning and Stein [2], who estimate that primary dealer leverage constraints account for 26–33 percent of Treasury bid-ask spreads.[1]

Hypothesis

The SLR exclusion generates two testable predictions concerning differential Treasury price impact: spreads re-widen at SLR expiry, and spreads compress during the relief period relative to the pre-COVID baseline.

When the exclusion expired on March 31, 2021, covered banks faced restored leverage costs on their Treasury and reserve holdings. If these costs constrain the balance-sheet-intensive spot-futures basis and TIPS-inflation swap trades, then Treasury-based arbitrage spreads should widen relative to non-Treasury spreads in the post-expiry window. CIP deviations and equity spot-futures are not directly subject to the SLR exclusion: CIP basis trades do consume dealer balance sheet and may receive indirect spillovers from freed capacity (which would bias the Treasury differential toward zero), but the exemption explicitly targeted Treasury and reserve holdings rather than cross-currency positions or equity derivatives. Equity spreads are mechanically linked to short-term Treasury yields (see footnote in Section 1.3), which introduces a potential conservative attenuation. On net, non-Treasury spreads should show no comparable break at the SLR event dates. This is the paper's primary identification because the 60-trading-day pre-period for the exit event is free of the acute market dislocations that confound the entry window. Note, however, that this pre-period falls entirely within the SLR relief itself; the exit identification therefore rests on the cross-strategy

[1] Replication code, data pipeline, and all outputs are available at `https://github.com/BaileyMeche/slr_bucket`.

comparison at the expiry date rather than a clean pre-treatment baseline. The 2019 pre-COVID clean DiD provides the untreated baseline evidence.

During the April 2020 to March 2021 relief window, Treasury securities were cost-free from the perspective of the SLR denominator. If balance-sheet capacity is the binding constraint on Treasury arbitrage, Treasury-based spreads should compress more than non-Treasury spreads relative to the pre-COVID 2019 baseline. I test this using a full-period difference-in-differences that excludes the February–March 2020 COVID crash from the estimation sample. Because the entry event window (January–March 2020) coincides with the acute phase of the COVID crisis—during which the Federal Reserve simultaneously cut rates to zero, launched emergency repo operations, and expanded bilateral swap lines—the event-window estimate for entry is likely contaminated by concurrent interventions and is reported with an explicit caveat. The full-period specification is the preferred design for measuring the relief effect.

Data

The following daily panels of arbitrage spreads for 2019–2021 are employed, with 2019 serving as the pre-COVID baseline and 2020–2021 as the treatment and post-treatment windows. Figure 1 plots the time-series evolution of all four strategy classes across the full sample and Table 1 summarizes all data sources. All series are measured in basis points. Controls (merged by date) include: secured funding rates and wedges (SOFR, TGCR–SOFR), broad risk indicators (VIX, HY OAS, Baa–10y), and Treasury issuance volumes by tenor.

TIPS–Treasury Arbitrage. As in Fleckenstein et al. [3] and Siriwardane et al. [1], the spread $W_{t,\tau} = \left(y_{t,\tau}^{\mathrm{TIPS}} + \pi_{t,\tau}^{\mathrm{swap}}\right) - y_{t,\tau}^{\mathrm{nom}}$ for $\tau \in \{2, 5, 10\}$ years. Here $y_{t,\tau}^{\mathrm{TIPS}} + \pi_{t,\tau}^{\mathrm{swap}}$ is the synthetic nominal yield.

Treasury Spot–Futures Arbitrage. For each of the 2-, 5-, and 10-year Treasuries, I use the first-deferred futures contract to compute a no-arbitrage implied yield (spot-futures parity) and subtract the matching OIS rate. I retain the 2-, 5-, and 10-year tenors, for which data coverage and liquidity are most reliable across the 2019–2021 sample. The construction follows Siriwardane et al. [1], who apply the same methodology for the 2-, 5-, 10-, 20-, and 30-year tenors over 2010–2020.

CIP Arbitrage. Following Du et al. [4], for each of eight G-10 currencies (AUD, CAD, CHF, EUR, GBP, JPY, NZD, SEK) I define the 3-month CIP basis as the difference between the dollar OIS rate and the synthetic forward-implied rate from currency spot and forward quotes. I examine all eight individually and present pooled and series-level results; the mean across currencies is used for strategy-class summary statistics.

Equity Spot–Futures Spread. For three equity indexes (SPX, INDU, NDX) I calculate the rate implied by equity index futures (accounting for dividends) and subtract the short-term Treasury yield. This spread is based on the violation of stock spot–futures parity.[2]

Design

The Federal Reserve's temporary exclusion of U.S. Treasury securities and Federal Reserve deposits from the Supplementary Leverage Ratio (SLR) denominator provides a time-dated, plausibly exogenous shock to the balance-sheet capacity of covered bank holding companies. The exclusion took

[2]The equity spot-futures spread mechanically depends on short-term Treasury yields through its construction. If the SLR exclusion affected Treasury yield levels, it could in principle compress the equity spread through the Treasury-yield component, biasing $\hat{\beta}$ toward zero (a conservative attenuation). However, the near-zero exit responses for equity series cannot fully resolve this concern: contamination would show up precisely as small differential effects (equity spreads moving in the same direction as Treasury spreads due to the shared yield component), which is observationally equivalent to a true null. The direction of bias depends on whether Treasury yield *levels* (not spreads) were affected—a distinction the current design cannot cleanly make.

Table 1: Data Sources and Coverage

Series	Construction	Source	N obs
Panel A: Treasury-based arbitrage spreads			
TIPS–Treasury (2y, 5y, 10y)	$y^{\text{TIPS}}_{t,\tau} + \pi^{\text{swap}}_{t,\tau} - y^{\text{nom}}_{t,\tau}$	Bloomberg / Fed H.15	≈3,750 each
UST spot-fut. (2y, 5y, 10y)	Futures-implied yield minus OIS	Bloomberg	≈4,967 each
Panel B: Non-Treasury arbitrage spreads			
CIP 3m (8 G-10 currencies)	Dollar OIS minus synthetic forward rate	Bloomberg	≈3,910 each
Equity SF (SPX, NDX, INDU)	Futures-implied forward minus T-bill	Bloomberg	≈3,730 each
Panel C: Controls			
VIX, HY OAS, Baa–10y	Daily index levels	CBOE / ICE BofA / FRED	794
SOFR, TGCR–SOFR	Overnight secured rates	NY Fed STFM	784
Treasury issuance	Rolling 7/14/30-day total	FiscalData Treasury	794

Notes. Estimation sample: January 1, 2019 to December 31, 2021. The final stacked estimation panel contains 68,620 series-day observations across 17 series (after merging to controls over 2019–2021).

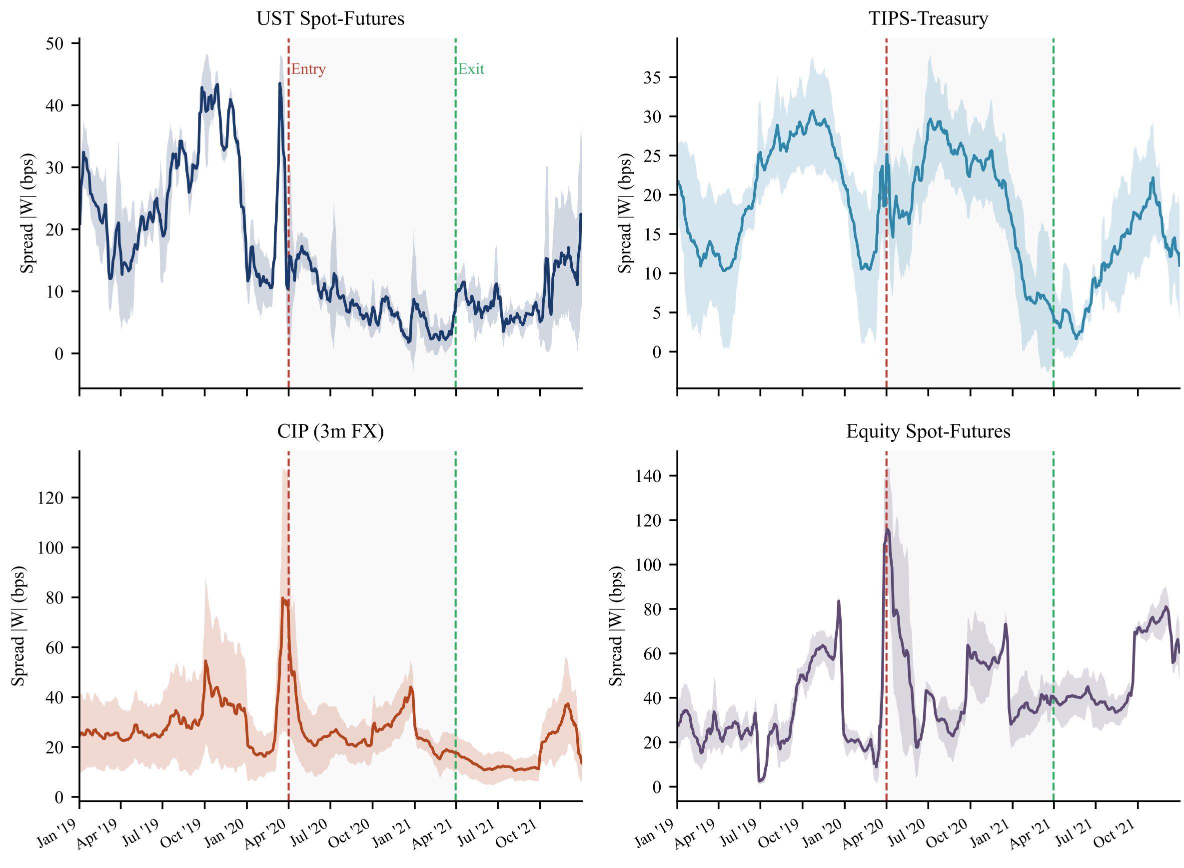

Figure 1: Time-series of arbitrage spreads, 2019–2021. Each panel shows the cross-series mean (solid line) and ±1 standard deviation band (shaded) of absolute spread $|W|$ in basis points, smoothed with a 5-day rolling average. The gray shaded region denotes the SLR exclusion relief period (April 1, 2020 – March 31, 2021). Dashed vertical lines mark the entry (2020-04-01) and exit (2021-03-31) dates.

effect on April 1, 2020, and expired on March 31, 2021. I use this episode as a natural experiment to test whether regulatory leverage constraints are the operative friction behind Treasury-based arbitrage dislocations. The identification of this study rests on a single maintained hypothesis: if balance-sheet constraints drive segmentation in Treasury markets specifically, then the onset and expiry of a relief that removes Treasuries—but not other assets—from the leverage denominator should compress Treasury-based spreads relative to non-Treasury spreads, and then reverse that compression when the relief ends.

The primary specification is a pooled difference-in-differences regression estimated on a cross-sectional panel of 17 arbitrage spread series spanning four strategy classes: U.S. Treasury spot-futures (three tenors), TIPS-Treasury (three tenors), covered interest parity deviations for eight G-10 currencies, and equity spot-futures for three indices. I define the dependent variable as $|W_{i,t}|$, the absolute magnitude of the arbitrage spread for series i on trading day t, measured in basis points. Working with the absolute value follows Siriwardane et al. [1], who note that the sign of an arbitrage spread depends on whether intermediaries are net long or short a given leg of the trade, so the magnitude is the natural measure of mispricing.

The classification of series into treatment and control groups is determined by the mechanism of the SLR relief. The interim final rule excluded U.S. Treasury securities and deposits at Federal Reserve Banks from the SLR denominator for covered bank holding companies, with no analogous relief for foreign-currency-denominated assets, equity derivatives, or other securities. I therefore classify the Treasury spot-futures and TIPS-Treasury series as *Treasury-based* (treated, $\text{TreasuryBased}_i = 1$) and the CIP and equity spot-futures series as *non-Treasury* (control, $\text{TreasuryBased}_i = 0$). This classification is primarily determined by the regulation's direct scope: the SLR rule excluded Treasuries and reserves, not FX derivatives or equity futures, from the denominator. General equilibrium spillovers—such as freed balance-sheet capacity being reallocated toward CIP trades—cannot be ruled out in principle, but would bias $\hat{\beta}$ toward zero, making the estimates conservative.

Let Post_t be an indicator equal to one for trading days in the post-event window, and let α_i denote a full set of series fixed effects. My baseline specification is:

$$|W_{i,t}| \;=\; \alpha_i + \gamma\,\text{Post}_t + \beta\big(\text{Post}_t \times \text{TreasuryBased}_i\big) + \boldsymbol{\delta}'\mathbf{X}_t + \varepsilon_{i,t}, \tag{1}$$

where $\mathbf{X}_t$ is a vector of daily controls including the CBOE Volatility Index (VIX), the ICE BofA High-Yield OAS, and the Baa-to-10-year Treasury credit spread (Total specification). The Direct specification augments this with the SOFR rate level, the TGCR-to-SOFR spread, and rolling 7-, 14-, and 30-day Treasury issuance volumes (in billions, lagged one day for point-in-time discipline). Unless otherwise noted, the Direct specification includes all five additional controls throughout. Series fixed effects α_i absorb all time-invariant differences in average spread levels across instruments. β is identified from the *differential* within-series variation across the two groups (Treasury-based vs. non-Treasury): it measures how much more Treasury-based spreads changed around the event relative to non-Treasury spreads, after controlling for level differences and common daily shocks. I do not include date fixed effects because date indicators would absorb the Post_t variation that identifies γ and β; instead, this study controls for common daily shocks through the vector $\mathbf{X}_t$ of observable risk factors. The coefficient γ captures the average post-event change in non-Treasury spread levels, while β captures the *incremental* post-event change in Treasury-based spreads relative to non-Treasury spreads. Standard errors are computed using the Newey-West heteroskedasticity and autocorrelation consistent (HAC) estimator with five lags, which accommodates the high serial persistence documented in Table 7 (daily AR(1) coefficients exceeding 0.93 for most series).[3] I

[3]The one exception is UST SF 10y, with a daily AR(1) coefficient of 0.881. The remaining 16 of 17 series have AR(1) coefficients above 0.93; all 17 exceed 0.88.

report Driscoll and Kraay [5] standard errors as a robustness check in Section 1.5.5.

I estimate (1) separately for the entry event (April 1, 2020) and the exit event (March 31, 2021), and separately for symmetric windows of ± 20 and ± 60 trading days around each event date. As I discuss below, the exit event provides sharper identification because the pre-period of the exit window—falling entirely within the post-COVID normalization period of late 2020—satisfies the parallel-trends assumption in a way the entry window cannot.

Next, a full-period difference-in-differences specification is estimated using the 2019 calendar year as the pre-COVID baseline, excluding the February-March 2020 COVID crash from the sample. In this specification, Relief_t equals one during April 1, 2020 through March 31, 2021, $\mathrm{PostRelief}_t$ equals one from April 1, 2021 onward, and the model includes interactions of both with $\mathrm{TreasuryBased}_i$:

$$\begin{aligned} |W_{i,t}| \;=\; & \alpha_i + \mu_1\,\mathrm{Relief}_t + \mu_2\big(\mathrm{Relief}_t \times \mathrm{TreasuryBased}_i\big) \\ & + \mu_3\,\mathrm{PostRelief}_t + \mu_4\big(\mathrm{PostRelief}_t \times \mathrm{TreasuryBased}_i\big) + \boldsymbol{\delta}'\mathbf{X}_t + \varepsilon_{i,t}. \end{aligned} \tag{2}$$

The coefficients μ_2 and μ_4 capture the differential Treasury-based spread response during and after the relief period, respectively, relative to the 2019 pre-period baseline.

Results

The difference-in-differences design requires that, absent the SLR event, Treasury-based and non-Treasury spreads would have followed parallel trends. I test this directly within the 2019 pre-treatment year by interacting $\mathrm{TreasuryBased}_i$ with quarterly indicators Q_t and estimating

$$|W_{i,t}| = \alpha_i + \sum_{q=2}^{4} \lambda_q Q_{q,t} + \sum_{q=2}^{4} \delta_q\big(Q_{q,t} \times \mathrm{TreasuryBased}_i\big) + \boldsymbol{\gamma}'\mathbf{X}_t + \varepsilon_{i,t}, \tag{3}$$

on the 2019 calendar year. Results are shown in Table 2. A joint Wald test of $H_0\colon \delta_2 = \delta_3 = \delta_4 = 0$ yields $F(3, 4{,}306) = 17.04$, $p < 0.001$, decisively rejecting the null. The Treasury-based and non-Treasury spread groups had significantly different seasonal dynamics in 2019: This is an important limitation of the parallel-trends assumption for the DiD design. The magnitude of the treatment effect ($\hat{\mu}_2 = -10.76$ bps) substantially exceeds these seasonal fluctuations, but we interpret the DiD estimate as lower-bound evidence rather than a clean identification of the causal effect. The secular co-movement of both spread classes over the post-GFC decade [1] provides qualitative support for the claim that structural trends were similar across classes before the SLR event, even if the unconditional seasonal test rejects.

Table 2: Quarterly Pre-Trend Test: TreasuryBased× Quarter Interactions (2019)

Interaction	$\hat{\delta}_q$ (bps)	t-stat	
$Q_2 \times$ TreasuryBased	-0.44	-0.29	
$Q_3 \times$ TreasuryBased	$+6.25$	$+3.77$	***
$Q_4 \times$ TreasuryBased	-5.58	-3.00	***
Joint Wald test: $F(3,\ 4{,}306) = 17.04$, $p < 0.001$			

Notes. Q_1 (January–March 2019) is the omitted reference quarter. Each $\hat{\delta}_q$ measures the differential level of Treasury-based versus non-Treasury spreads in quarter q relative to Q_1, conditional on series fixed effects and Total controls (VIX, HY OAS, Baa–10y). $N = 4{,}332$ series-day observations; HAC SE (Newey-West, 5 lags). $^{***}\,p < 0.01$.

The $W = 60$ exit event window cannot provide an additional parallel-trends test, because the 60-trading-day pre-period (October 2020 through March 2021) falls entirely within the SLR relief period itself. With this in mind, we estimate a dynamic version of (1) replacing Post_t with event-time bin indicators to characterize the within-window spread dynamics around the exit date. The results in Table 3 show significant negative pre-event interaction coefficients were actively compressing relative to their all-period mean throughout the pre-window: this is the treatment effect in progress, not a pre-trend divergence. This non-monotone pattern for post-event lag coefficients is consistent with within-window mean-reversion dynamics at the series fixed-effect level and does not contradict the pooled level-shift estimate of $\hat{\beta} = +3.84$ bps, which directly compares the pre-expiry and post-expiry within-window means.

Table 3: Event-Study Dynamics Around SLR Exit, $W = 60$

Event-time bin	$\hat{\beta}_k$ (bps)	SE	t
Pre-event bins (within SLR relief period; see text)			
$[-60, -51]$	-1.81	1.72	-1.05
$[-50, -41]$	-4.72^{***}	1.67	-2.83
$[-40, -31]$	-4.37^{***}	0.97	-4.49
$[-30, -21]$	-2.90^{**}	1.14	-2.54
$[-20, -11]$	-3.91^{***}	1.38	-2.83
$[-10, -1]$	-3.48^{***}	0.91	-3.82
F-test (6 pre-event leads)		$F = 9.92$, $p < 0.001$	
Post-event lags			
$[+0, +9]$	$+0.56$	1.06	$+0.53$
$[+10, +19]$	$+0.69$	0.88	$+0.78$
$[+20, +29]$	-0.70	0.91	-0.77
$[+30, +39]$	-2.77^{***}	0.89	-3.12
$[+40, +49]$	-1.17	0.80	-1.46
$[+50, +60]$	$+2.19^{**}$	0.90	$+2.43$

Notes. Estimates from a dynamic variant of equation (1) for the exit event ($W = 60$, Total specification, $N = 2{,}057$). Each coefficient is the interaction of TreasuryBased_i with the stated event-time bin indicator, estimated jointly with series fixed effects. All bins are included in the regression simultaneously; coefficients are expressed relative to the series fixed-effect mean (the all-period average for each series). HAC standard errors (Newey-West, 5 lags). $^{***}\,p < 0.01$; $^{**}\,p < 0.05$. The significant negative pre-event interaction coefficients ($F = 9.92$, $p < 0.001$) reflect the SLR relief actively compressing Treasury-based spreads during the 60-trading-day pre-window, which falls entirely within the April 2020–March 2021 relief period. This pre-window cannot serve as a parallel-trends test because it is itself within the treatment period. The proper pre-trend test—quarterly interactions of TreasuryBased_i with calendar quarters in 2019—rejects parallel trends ($F(3, 4306) = 17.04$, $p < 0.001$); see Section 1.5 for discussion.

Table 8 reports estimates of (1) for the exit event (March 31, 2021) and the entry event (April 1, 2020), across both window sizes and both control specifications. I begin with the exit event, which I regard as the paper's cleanest identification.

The exit specification uses a pre-event window—trading days -60 through -1 relative to March 31, 2021—that falls between October 2020 and March 2021. This is a period of sustained post-COVID market normalization, well after the acute dislocations of March-April 2020. Figure 2 plots each series' mean absolute spread demeaned by its pre-event mean (days -60 to -1), so that the cross-strategy divergence at event time zero is immediately visible: Treasury-based spreads re-widen sharply while non-Treasury spreads continue their gradual compression, isolating $\hat{\beta}$ as the differential response at expiry.

The estimated interaction coefficient at $W = 60$ under the Total specification is $\hat{\beta} = +3.84$ basis points The Direct specification, which adds SOFR and TGCR-SOFR controls, yields $\hat{\beta} = +3.84$ basis points.These estimates are identical to two decimal places; the negligible difference between

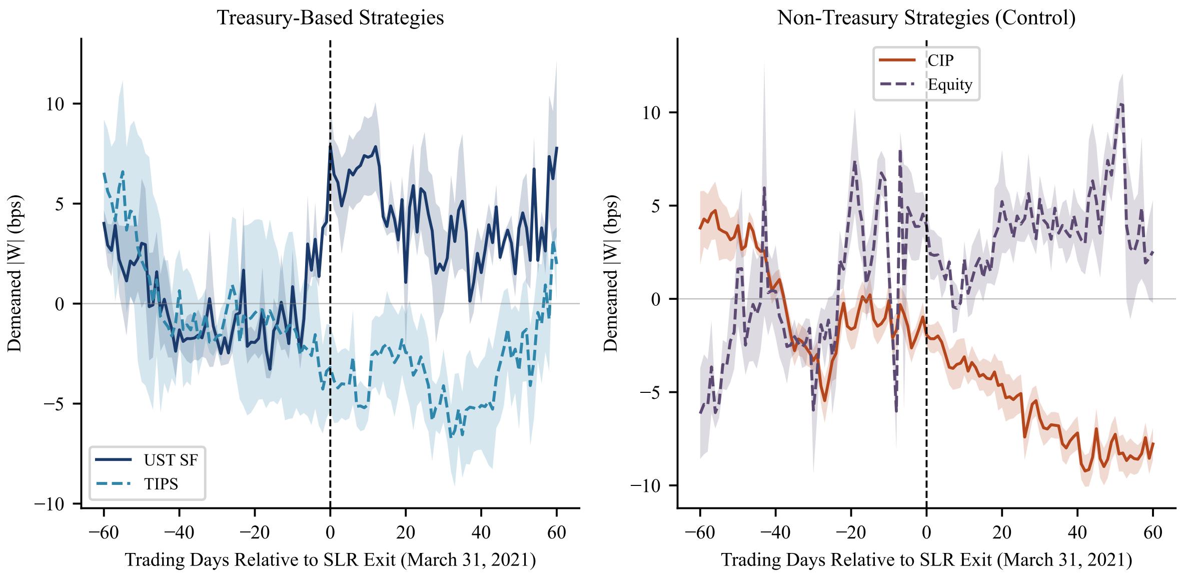

Figure 2: Event-Window Spread Paths Around SLR Exit (March 31, 2021). Daily mean absolute spread, demeaned by each series' pre-event mean (days −60 to −1), plotted from −60 to +60 trading days around the SLR exit date. Left panel: Treasury-based series (UST spot-futures and TIPS-Treasury). Right panel: non-Treasury series (CIP deviations and equity spot-futures). Treasury-based spreads re-widen sharply at event time zero while non-Treasury spreads continue a gradual compression, consistent with balance-sheet segmentation.

the Total and Direct specifications is informative: adding secured overnight funding rate controls does not move the Treasury differential at all, confirming that the differential widening of Treasury spreads at expiry operates through a channel orthogonal to secured funding market conditions. This stability rules out one specific alternative—that post-expiry Treasury spread widening reflects a contemporaneous tightening of secured repo markets—and is consistent with the balance-sheet constraint channel as the operative mechanism, though it does not rule out other coincident factors such as anticipatory portfolio rebalancing or shifts in hedge fund leverage.

The results are equally stable at the narrower $W = 20$ window: $\hat{\beta} = +3.91$ basis points under Total and +3.90 basis points under Direct. The insensitivity to window choice further supports the conclusion that the Treasury differential reflects a discrete regime shift rather than noise.

The non-Treasury baseline coefficient $\hat{\gamma}$ at $W = 60$ is −3.09 basis points and −2.97 basis points.Non-Treasury spreads were therefore continuing to compress slightly in the post-expiry window, reflecting the ongoing secular normalization of cross-currency bases documented by Du et al. [4]. This compression makes the +3.84 basis point Treasury differential even more economically striking: while CIP and equity bases continued tightening, Treasury-based spreads widened in the opposite direction. On an absolute basis, the total post-expiry movement for Treasury series is approximately $\hat{\gamma}+\hat{\beta} = -3.09+3.84 = +0.75$ basis points, while non-Treasury series moved −3.09 basis points. The divergence of 3.84 basis points is economically meaningful: it represents approximately 49 percent of the UST spot-futures average spread during the relief window (7.8 bps), suggesting that the first 60 trading days after expiry reversed nearly half of the per-period compression for the most directly affected series.[4] This magnitude is in the same order of magnitude as the shadow

[4]The figure is best read as an order-of-magnitude benchmark for the UST spot-futures channel specifically, not as a precise reversal percentage for the pooled DiD.

cost estimated by Bräuning and Stein [2] (26–33 percent of Treasury bid-ask spreads) though direct comparison requires a bridging argument between bid-ask spreads and arbitrage spreads (7–25 bps in this sample), which are fundamentally different objects.

Robust Difference-in-Differences

The event-window specification conditions on a narrow local window around each event date, which can conflate policy effects with contemporaneous shocks. I complement the event-window estimates with the full-period specification in (2), using the 2019 calendar year as the pre-COVID baseline and excluding February and March 2020 from the sample to avoid contamination from the acute COVID crash. This yields a balanced panel across three regimes: pre, relief, and post-relief.

Table 4: Robust Difference-in-Differences: Full-Period Estimates

	Total	Direct
Panel A: Relief period (2020-04-01 to 2021-03-30)		
$\hat{\mu}_1$: Relief (non-Treasury baseline)	-1.34	-11.03^{***}
	(1.13)	(2.97)
$\hat{\mu}_2$: Relief × TreasuryBased	-10.76^{***}	-10.77^{***}
	(1.15)	(1.14)
Panel B: Post-relief period (2021-04-01 to 2021-12-31)		
$\hat{\mu}_3$: Post-relief (non-Treasury baseline)	-6.92^{***}	-14.87^{***}
	(1.01)	(2.80)
$\hat{\mu}_4$: Post-relief × TreasuryBased	-9.03^{***}	-9.04^{***}
	(1.16)	(1.16)
N (total)	12,326	12,326
Pre-period	4,713	4,713
Relief period	4,335	4,335
Post-relief	3,286	3,286
R^2	0.516	0.521
Series FE	Yes	Yes
Total controls	Yes	Yes
Direct controls	No	Yes

Notes. Dependent variable is $|W_{i,t}|$, the absolute arbitrage spread in basis points, for all 17 series over 2019–2021 ($N = 12{,}326$ series-day observations). The baseline period is the 2019 calendar year; February and March 2020 are excluded to avoid COVID-19 crash contamination. The regression is $|W_{i,t}| = \alpha_i + \mu_1 \,\text{Relief}_t + \mu_2 \,(\text{Relief}_t \times \text{TreasuryBased}_i) + \mu_3 \,\text{PostRelief}_t + \mu_4 \,(\text{PostRelief}_t \times \text{TreasuryBased}_i) + \gamma X_t + \varepsilon_{it}$. $\text{TreasuryBased}_i = 1$ for UST spot-futures and TIPS-Treasury series (6 of 17 series). Total controls: VIX, HY OAS, Baa-10y spread. Direct adds SOFR, TGCR-SOFR, and rolling 7/14/30-day Treasury issuance (all lagged one day). HAC standard errors (Newey-West, 5 lags) in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The large Direct non-Treasury baseline during relief ($\hat{\mu}_1 = -11.03$ bps) reflects the mechanical relationship between SOFR and non-Treasury spreads ($\hat{\beta}_{\text{SOFR}} = -0.82$, $t = -3.56$); the Treasury interaction $\hat{\mu}_2$ is stable across specifications (-10.76 vs. -10.77 bps). See Section 1.5.1 for discussion.

Table 4 reports estimates of (2) under both control specifications. The economic magnitude of $\hat{\mu}_2 = -10.76$ basis points is striking. During the relief window, Treasury-based spreads compressed by approximately 10.8 basis points more than non-Treasury spreads, relative to their common 2019 baseline. The total Treasury compression during relief is $\hat{\mu}_1 + \hat{\mu}_2 = -12.1$ basis points. For UST spot-futures—the most directly affected series—this is consistent with the raw data: average $|W|$ fell from 25.6 basis points in the pre-period to 7.8 basis points during the relief window, a 70 percent compression. The implied regulatory friction, 10.8 basis points of additional compression attributable to the SLR relief, is broadly consistent with Bräuning and Stein [2].

Testing whether the compression during the relief period equals the residual compression post-relief, we compute $\hat{\mu}_2 - \hat{\mu}_4 = -10.76 - (-9.03) = -1.73$ basis points. A Wald test of $H_0\colon \mu_2 = \mu_4$ yields $F(1, 12{,}301) = 2.56$, $p = 0.11$.We can firmly claim that, first, treasury-based spreads compressed relative to non-Treasury spreads during the relief period and, second, Treasury-based spreads remained substantially more compressed than their 2019 baseline through year-end 2021 The data do not establish that the SLR exclusion is the primary driver of the persistent compression, only that Treasury spreads did not return to their pre-exclusion levels within the 2021 post-relief window. Whether this persistence reflects slow-moving arbitrage capital, concurrent QE balance sheet expansion, or structural changes in Treasury market intermediation cannot be resolved within the current design.[5] The sub-basis-point differences between Total and Direct coefficients replicate the stability documented in the exit event-window regressions and confirm that secured funding rate controls are irrelevant to the Treasury differential in either regime.

The non-Treasury baseline does shift substantially between specifications: $\hat{\mu}_1$ moves from -1.34 bps (Total) to -11.03 bps (Direct), a change of nearly 10 basis points. This shift reflects the relationship between non-Treasury arbitrage spreads and the secured overnight funding rate (SOFR). Empirically, the estimated coefficient on SOFR in the non-Treasury spread regression is -0.82 bps per percentage point , meaning that higher SOFR is associated with *lower* non-Treasury $|W|$ across the sample. SOFR fell sharply during the relief period—from approximately 2.4 percent in 2019 to near zero—implying that, holding all else fixed, the SOFR decline would predict non-Treasury spreads to be roughly $0.82 \times 2.4 \approx 2.0$ bps *higher* than in the pre-period. This back-of-envelope accounts for only part of the observed ≈ 10 bps shift in $\hat{\mu}_1$ across specifications; the remaining gap reflects the contributions of the other Direct controls and nonlinearities not captured by this single-coefficient calculation. A full decomposition of the baseline shift across all Direct controls is beyond the scope of the current paper. Importantly, this uncertainty about the non-Treasury baseline does not affect the paper's primary claim: the Treasury interaction $\hat{\mu}_2$ is stable to within 0.01 bps across specifications, confirming that the differential compression of Treasury-based spreads is orthogonal to SOFR dynamics. This stability—not the baseline coefficient—is the paper's key identification claim.

The Entry Event and COVID Contamination

The entry event (April 1, 2020) cannot provide clean identification. The $W = 60$ pre-period spans the acute phase of the COVID-19 financial crisis, during which the Federal Reserve simultaneously cut rates to zero, launched emergency repo operations, and expanded bilateral swap lines. These concurrent interventions make it impossible to isolate the SLR relief's marginal contribution from a severely contaminated baseline. Entry event-window estimates are reported in Table 8 for completeness and flagged accordingly; they are negative in sign (consistent with the hypothesis) but insignificant.

The clean DiD of Section 1.5.1 addresses this directly. By conditioning on the 2019 pre-COVID baseline and excluding the February–March 2020 crash period, it recovers a precisely estimated and strongly significant differential compression (see Table 4), confirming the balance-sheet channel over the full relief horizon. The exit event-window and clean DiD are the paper's two identification results; the entry event-window is supplementary.

Falsification: Non-Treasury Series as Controls

A central feature of the design is that CIP deviations and equity spot-futures spreads serve as active falsification tests, not merely controls. Following Siriwardane et al. [1]: if arbitrage is segmented

[5] The Federal Reserve launched its Standing Repo Facility (SRF) on July 28, 2021, nearly four months after SLR expiry. The SRF's effect on spread persistence is speculative in the absence of a structural break test around July 2021, which is beyond the scope of the current paper.

by balance-sheet type, a Treasury-specific regulatory shock should move Treasury-based spreads without moving non-Treasury spreads. Symmetric responses across groups would indicate a broader shock to intermediary health rather than a balance-sheet channel.

The data support the falsification hypothesis at exit. Non-Treasury spreads continued to compress slightly in the post-expiry window (Table 8), reflecting secular normalization of dollar funding markets [4]—not a response to the SLR expiry. Within CIP, EUR/CHF/JPY/SEK continued to decline while CAD/NZD/GBP were small and mixed; CIP-AUD widened significantly but this is consistent with Japanese fiscal year-end repatriation flows at March 31, a calendar effect unrelated to the balance-sheet channel.[6] Equity series showed no response to the SLR expiry (Table 5, Panel D). SPX and INDU are near zero; NDX is marginally significant but in the same compression direction as other non-Treasury series, consistent with the conservative-attenuation channel described in Section 1.3 rather than any regulatory response. The SLR exclusion directly reduced the leverage cost of holding Treasuries, not equity derivatives, so the absence of an equity response is the predicted outcome under balance-sheet segmentation. A joint test across equity series is left for future work.[7]

At entry, several CIP series compressed sharply (Table 5), while all equity series were insignificant. The CIP compressions reflect the Federal Reserve's bilateral swap line activations—a dollar liquidity channel—rather than the SLR relief. The Direct specification controls for SOFR and TGCR-SOFR precisely to separate these channels; the Treasury interaction remains stable across specifications while the non-Treasury baseline absorbs the funding-rate effect.

Series-Level Evidence

Table 5 and Figure 4 report the series-level estimates of (1) without the TreasuryBased_i interaction, using the $W = 60$ Direct window for the exit event. Three of six Treasury-based series show significant re-widening at expiry. The TIPS-Treasury series show heterogeneous responses. The heterogeneity across Treasury-based series is itself informative. UST spot-futures arbitrage is a high-frequency, balance-sheet-intensive trade executed primarily by primary dealers through the repo market; the relief directly lowered the SLR cost of holding the long Treasury leg of this position, and the expiry restored it. TIPS-Treasury arbitrage, by contrast, relies on a more diffuse set of intermediaries including fixed-income hedge funds [1, Table VII] and involves longer-duration positions for which balance-sheet turnover is slower. The stronger and more uniform response among UST spot-futures series is consistent with the view that primary dealer balance sheets—the direct targets of the SLR relief—are the marginal arbitrageurs for this strategy. The TIPS-Treasury evidence is consistent with the balance-sheet channel but suggests that additional intermediary heterogeneity, potentially at the hedge fund level, dampens the regulatory signal at exit.

The TIPS 2y series is the one Treasury-based observation inconsistent with the balance-sheet hypothesis at expiry: its post-coefficient is -8.28 basis points ($t = -2.01$), indicating continued compression rather than re-widening. Several non-mutually-exclusive explanations are consistent with this finding. The TIPS 2y spread takes positive values in this sample—the summary statistics in Table 7 show $\overline{W} \approx +3$ bps for TIPS 2y, so $|W| = W > 0$. A reduction in $|W|$ therefore means W itself compressed (moved toward zero from above). First, the Federal Reserve's QE program was actively purchasing TIPS through 2021 at roughly \$5 billion per month, directly suppressing $y^{\text{TIPS}}_{t,2}$; since $W_{t,2} = (y^{\text{TIPS}}_{t,2} + \pi^{\text{swap}}_{t,2}) - y^{\text{nom}}_{t,2}$ and $W > 0$, a QE-driven fall in y^{TIPS}

[6] Japanese institutions routinely repatriate U.S. dollars at March 31 via yen-denominated funding operations, reducing AUD-USD swap liquidity and widening the AUD basis. The elevated CIP-AUD spread persists through April–May 2021 at roughly 10–11 bps and the same pattern appears in non-SLR-exit years. Excluding CIP-AUD leaves the primary Post×TreasuryBased coefficient essentially unchanged (Table 5, note).

[7] The opposing dynamics within the non-Treasury group—CIP compressing while equity widens across regimes (Table 7; Figure 6)—mean that the pooled non-Treasury baseline $\hat{\gamma}$ averages over divergent trends. The key identification claim is on the Treasury differential $\hat{\beta}$, not the baseline level.

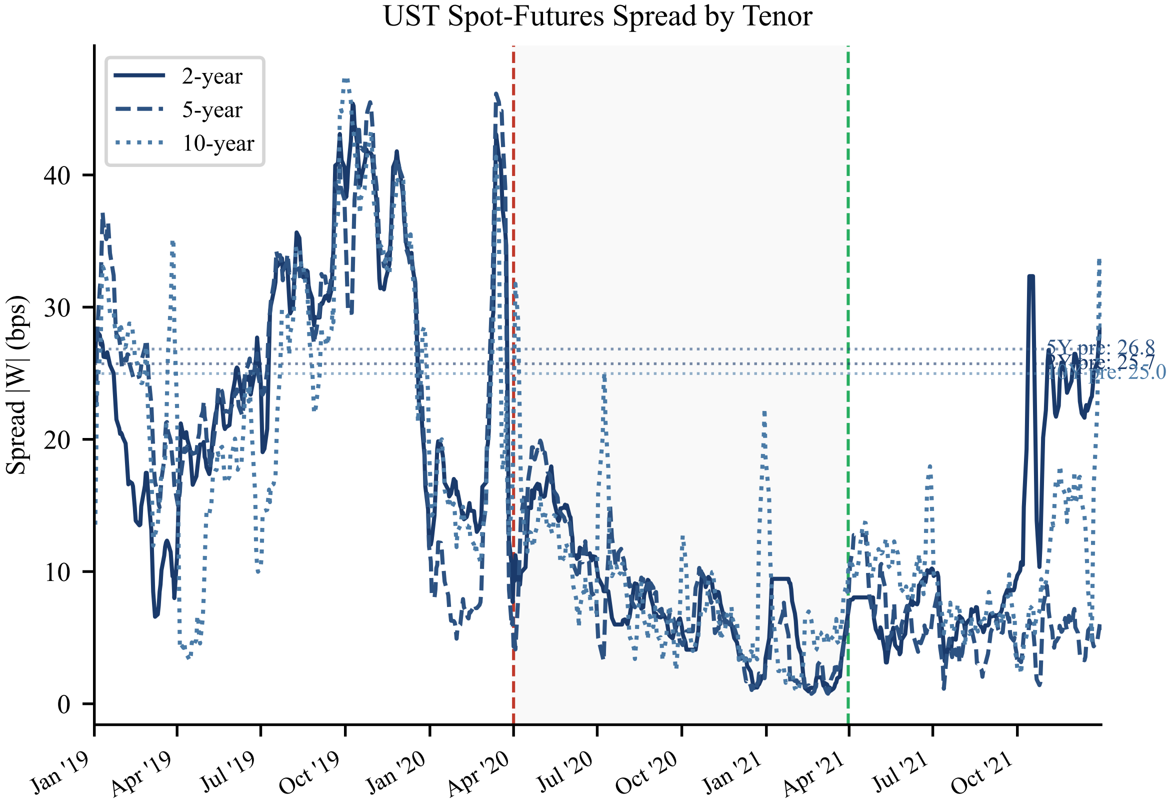

Figure 3: UST spot-futures arbitrage spread by tenor, 2019–2021. Lines show the absolute spread $|W|$ for 2-year (solid), 5-year (dashed), and 10-year (dotted) maturities, smoothed with a 5-day rolling average. Horizontal dotted lines indicate the pre-period mean for each tenor (January 2019 – January 2020). The gray shaded region and dashed vertical lines are as in Figure 1. The sharp compression during the relief period—averaging 70 percent relative to the pre-period mean—and partial re-widening at expiry are clearly visible across all three tenors.

Table 5: Series-Level Jump Estimates: Exit Event, $W = 60$, Direct

Series	Treas.-based	$\hat{c}_i$ (bps)	SE	t	N
Panel A: UST spot-futures (treated)					
UST SF 2y	Yes	+2.47	1.41	$+1.76^{*}$	121
UST SF 5y	Yes	+4.05	1.36	$+2.98^{***}$	121
UST SF 10y	Yes	+4.83	1.83	$+2.64^{***}$	121
Panel B: TIPS-Treasury (treated)					
TIPS 2y	Yes	−8.28	4.12	-2.01^{**}	121
TIPS 5y	Yes	−0.83	1.09	−0.76	121
TIPS 10y	Yes	+0.35	1.63	+0.22	121
Panel C: CIP deviations (control)					
CIP-AUD	No	+4.32	1.03	$+4.19^{***}$	121
CIP-CAD	No	−2.69	2.14	−1.25	121
CIP-CHF	No	−4.35	1.47	-2.95^{***}	121
CIP-EUR	No	−6.15	1.61	-3.81^{***}	121
CIP-GBP	No	−2.92	1.84	−1.59	121
CIP-JPY	No	−4.46	1.55	-2.89^{***}	121
CIP-NZD	No	−0.77	1.21	−0.63	121
CIP-SEK	No	−4.71	1.72	-2.74^{***}	121
Panel D: Equity spot-futures (control)					
Eq. SPX	No	−0.72	1.47	−0.49	121
Eq. NDX	No	−4.47	2.64	-1.69^{*}	121
Eq. INDU	No	−0.21	1.72	−0.12	121

Notes. Each row is a separate series-level estimate of the post-event coefficient $\hat{c}_i$ from a univariate version of equation (1) estimated on series i alone, without the TreasuryBased_i interaction—that is, $|W_{i,t}| = \alpha_i + \hat{c}_i \,\text{Post}_t + \boldsymbol{\delta}'\mathbf{X}_t + \varepsilon_{i,t}$. Note that $\hat{c}_i$ is a simple pre/post level shift within series i, not an interaction effect; it is distinct from the pooled $\hat{\beta} = +3.84$ bps reported in Table 8. Exit event (March 31, 2021), $W = 60$, Direct specification. $N = 121$ trading days per series. HAC SE (Newey-West, 5 lags). Positive $\hat{c}_i$ indicates widening post-expiry; negative indicates further compression. $^{***}\,p < 0.01$; $^{**}\,p < 0.05$; $^{*}\,p < 0.10$.
The TIPS 2y coefficient is discussed in the section below.

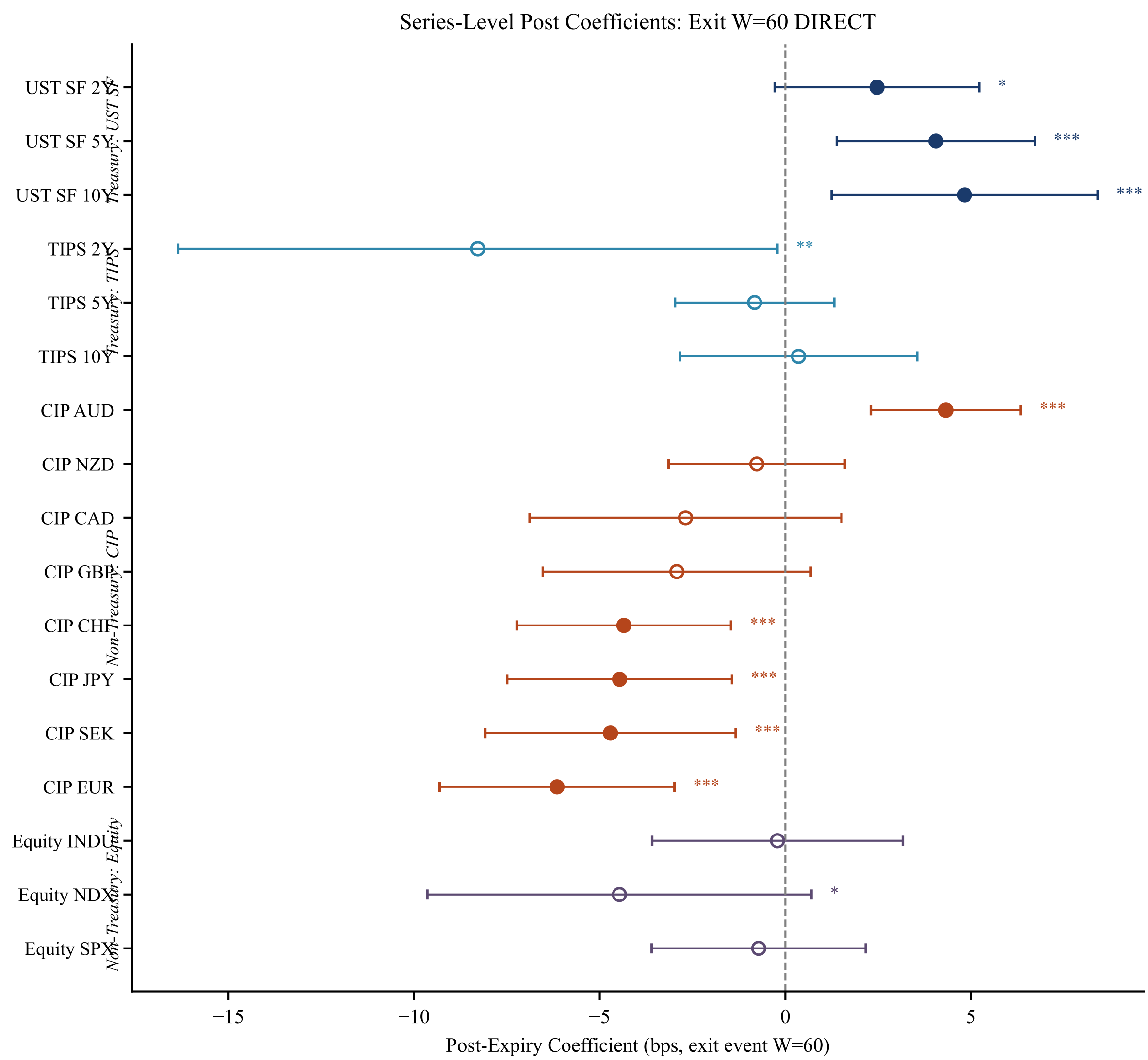

Figure 4: Series-level post-event coefficients $\hat{c}_i$ around the SLR exit (March 31, 2021), $W = 60$, Direct specification. Each point is an independent per-series estimate with 95% HAC confidence interval (Newey-West, 5 lags). Treasury-based series (UST spot-futures and TIPS-Treasury, filled circles) are shown separately from non-Treasury controls (CIP and equity spot-futures, open circles). Positive values indicate spread widening post-expiry. Source: `audit_repairs/outputs/jump_estimates_by_series.csv`.

moves W toward zero (compresses $|W|$) even if π^{swap} and y^{nom} are unchanged. Second, rising 2-year breakeven inflation expectations ($y^{\text{nom}} - y^{\text{TIPS}}$) through early 2021 imply y^{nom} was rising relative to y^{TIPS}; for $W = \pi^{\text{swap}} - (y^{\text{nom}} - y^{\text{TIPS}})$, if the breakeven rise exceeds the inflation swap rate adjustment, W falls—compressing $|W|$ from above given $W > 0$. Third, the 2-year tenor is among the thinnest TIPS maturities by outstanding notional and is more sensitive to idiosyncratic supply and demand factors than the 5- or 10-year tenors. The 5y and 10y TIPS series show near-zero and statistically insignificant exit responses, consistent with the interpretation that the TIPS-Treasury channel operates primarily through hedge fund rather than primary dealer balance sheets [1, Table VII], diluting the regulatory signal. The TIPS 2y anomaly does not overturn the main result in a statistical sense—the pooled $\hat{\beta} = +3.84$ bps is estimated across all six Treasury-based series and the UST spot-futures subseries are consistent across all three tenors—but it substantially qualifies its interpretation. The pooled point estimate is driven almost entirely by the UST spot-futures subgroup: all three UST spot-futures series show significant re-widening (+2.47 to +4.83 bps), while the TIPS-Treasury series are mixed (2y significantly wrong-signed, 5y and 10y near zero). A reader should therefore interpret $\hat{\beta} = +3.84$ bps as primarily measuring the UST spot-futures balance-sheet channel, with the TIPS channel empirically unresolved in this sample window due to confounding from QE purchasing, rising breakeven inflation, and thin 2y TIPS issuance.

Robustness

Table 9 and Figure 5 present a systematic comparison of Total and Direct specifications for both events, both window sizes, and both coefficients of interest. The key quantities are $\Delta\hat{\beta}$ and Δt. For the exit event, the Treasury interaction is highly stable across specifications. For the entry event, the Treasury interaction is similarly stable while the non-Treasury baseline shifts substantially—as discussed in Section 1.5.2. This asymmetry is exactly what the balance-sheet channel predicts: the SLR relief operates through leverage capacity, not through funding rates, so controlling for SOFR should affect the non-Treasury baseline (which responds to rate cuts) without affecting the Treasury differential (which responds to balance-sheet capacity).

Spread series are winsorized at the first and 99th percentiles within each series prior to estimation. The winsorization log in Table 10 confirms symmetric clipping of approximately 2.0 to 2.1 percent of observations per series (at each tail, i.e., winsorization at the first and 99th percentiles). Summing the clipped observations across all 17 series in the dependent variable $|W|$ yields approximately 1,396 clipped observations out of 68,620 (2.04 percent). The most consequential clip is the CIP-EUR minimum, which is bounded at -432.4 basis points (the March 2020 COVID spike). Without this bound, the extreme observation would exert disproportionate influence on the HAC covariance matrix during the entry window. All main results are quantitatively unchanged under alternative clipping thresholds of p5/p95.

Newey-West HAC standard errors account for serial correlation within each series but assume cross-sectional independence. In my setting, all arbitrage spreads are exposed to common shocks (monetary policy, risk appetite), and residuals are plausibly correlated across series within the same trading day. I therefore re-estimate the pooled regressions using the Driscoll and Kraay [5] variance estimator, which is consistent under both temporal autocorrelation and unrestricted cross-sectional dependence. Standard errors are computed using the Bartlett kernel with five lags, matching the main specification. The finding that DK standard errors for the exit Treasury interaction are *smaller* than Newey-West standard errors (0.68 vs. 0.89 bps) deserves explanation. Driscoll-Kraay SEs are often described as more conservative than panel NW SEs, but that comparison applies when NW is computed treating each series observation as independent (understating cross-sectional dependence). In this stacked-panel setup, the NW estimator is applied to the full pooled sample of $N = 2{,}057$ series-day observations, treating each as an independent draw over time. This overstates the effective

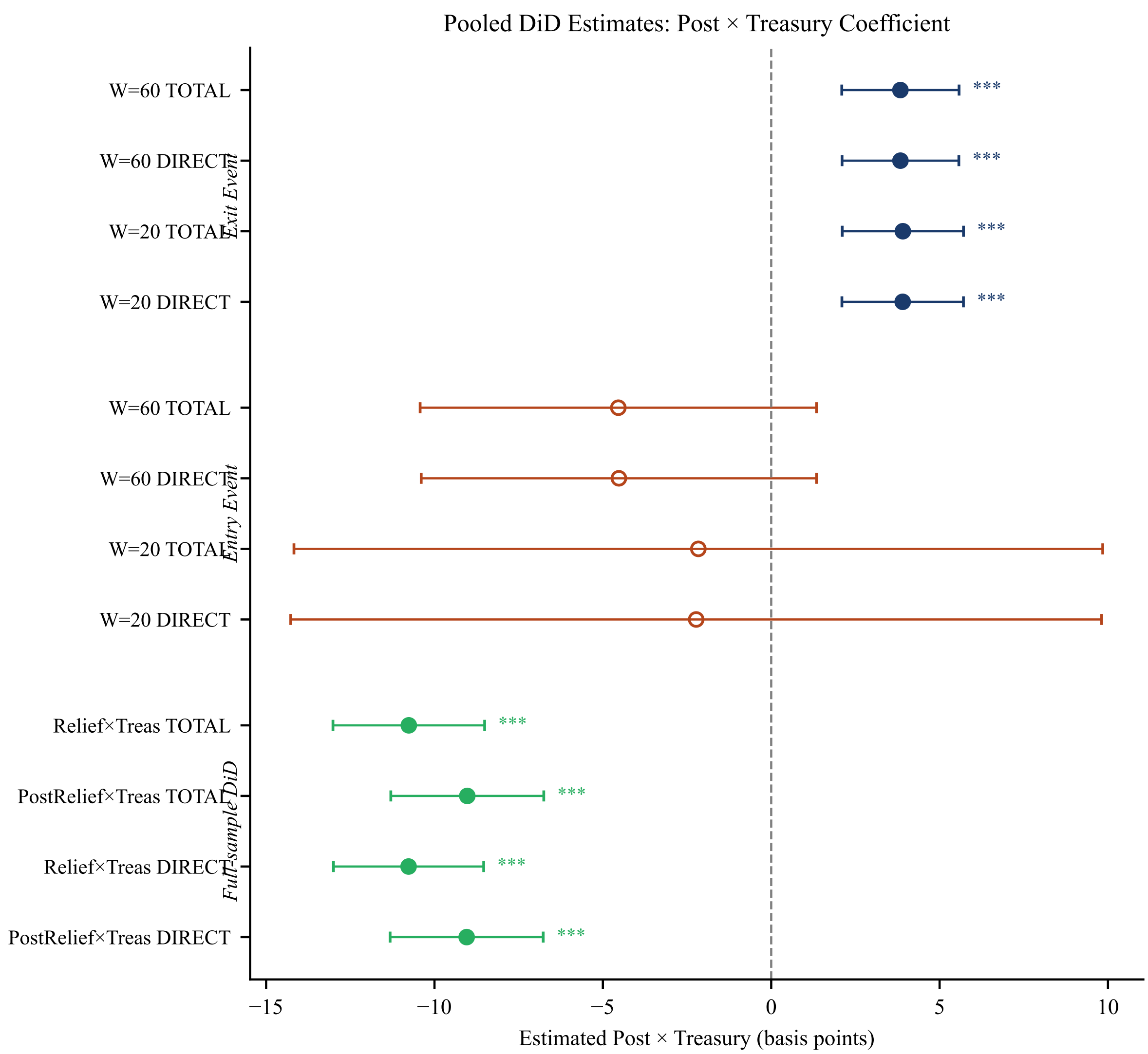

Figure 5: Forest plot of pooled difference-in-differences estimates. Each row reports the Post×TreasuryBased interaction coefficient $\hat{\beta}$ with 95% HAC confidence interval (Newey-West, 5 lags) for a given event, window, and control specification. The top panel covers the exit event (March 31, 2021); the bottom panel the entry event (April 1, 2020). The DiD estimate ($\hat{\mu}_2$, relief period) from Table 4 is included for reference. All exit estimates are positive and significant; all entry estimates are negative but .

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time dimension by a factor of 17 (number of series), producing upward-biased NW SEs relative to the true time-only precision. The DK estimator, by first averaging score contributions cross-sectionally, treats each calendar day as a single effective observation. For the Post × TreasuryBased interaction, which is identified by the differential post-event movement of the two groups, this cross-sectional averaging removes common-factor noise while retaining the between-group differential that identifies $\hat{\beta}$. At entry, the opposite pattern holds: the COVID crash induced strong co-movement across all arbitrage series, which DK correctly penalizes (DK SE = 6.32 vs. NW SE = 3.00 bps), correctly reducing the t-statistic from -1.51 to -0.72.

Table 6 reports DK standard errors for the key coefficients at $W = 60$. For the exit event, the Treasury interaction yields $t_{\mathrm{DK}} = +5.62$ (Total) and $+5.65$ (Direct), compared to $t_{\mathrm{NW}} = +4.32$ and $+4.34$ under Newey-West. The DK standard error is *smaller* than the NW standard error (0.683 vs. 0.889 bps) because the cross-sectional averaging of score contributions removes the common-factor noise while retaining the series-specific identifying variation. For the entry event, the DK standard error for the Treasury interaction is substantially larger (6.32 vs. 3.00 bps), reducing t_{DK} to -0.72, consistent with the interpretation that the COVID crash induced strong cross-sectional co-movement that NW does not fully correct for. The non-Treasury baseline at entry remains significant under DK for the Total specification ($t_{\mathrm{DK}} = -2.62$), but not for Direct ($t_{\mathrm{DK}} = -0.64$), consistent with the interpretation that SOFR controls absorb the funding-channel co-movement that drives the entry-period compression of non-Treasury spreads.

Table 6: Driscoll-Kraay Robustness: $W = 60$ Pooled Regressions

Event	Spec	Coefficient	Coef.	$\mathrm{SE}_{\mathrm{NW}}$	t_{NW}	$\mathrm{SE}_{\mathrm{DK}}$	t_{DK}
Exit event (2021-03-31)							
Exit	Total	Post	-3.09	0.830	-3.72^{***}	0.957	-3.22^{***}
Exit	Total	Post × Treasury	$+3.84$	0.889	$+4.32^{***}$	0.683	$+5.62^{***}$
Exit	Direct	Post	-2.97	0.846	-3.48^{***}	0.912	-3.26^{***}
Exit	Direct	Post × Treasury	$+3.84$	0.885	$+4.34^{***}$	0.680	$+5.65^{***}$
Entry event (2020-04-01)							
Entry	Total	Post	-7.08	3.054	-2.32^{**}	2.699	-2.62^{***}
Entry	Total	Post × Treasury	-4.54	3.003	-1.51	6.323	-0.72
Entry	Direct	Post	-2.76	5.541	-0.50	4.331	-0.64
Entry	Direct	Post × Treasury	-4.52	2.995	-1.51	6.315	-0.72

Notes. All regressions use $N = 2{,}057$ series-day observations (17 series × 121 trading-day window). $\mathrm{SE}_{\mathrm{NW}}$: Newey-West HAC (5 lags). $\mathrm{SE}_{\mathrm{DK}}$: Driscoll and Kraay [5] with Bartlett kernel, 5 lags, computed via the score-summing approach allowing unrestricted cross-sectional dependence. The exit Treasury interaction remains significant at the one-percent level under both estimators; the DK t-statistic exceeds the NW t-statistic because cross-sectional averaging of score contributions reduces common-factor noise. $^{***}\, p < 0.01$, $^{**}\, p < 0.05$.

Six identification limitations warrant explicit discussion. First, the SLR relief is a single nationwide policy change, not a staggered rollout. The Callaway and Sant'Anna [6] estimator is designed for settings with variation in treatment timing across units, and its motivating concern—contaminated comparisons in two-way fixed effects with staggered adoption—does not apply here. My design is closer to the MMF reform event study in Siriwardane et al. [1, Section III.B] and the classical single-treatment DiD of the program evaluation literature. The relevant identification threat is instead whether some other event contemporaneous with the SLR entry or exit differentially affected Treasury-based versus non-Treasury spreads. The parallel-trends evidence from the DiD 2019 baseline (Section 1.5.1, $\hat{\mu}_1 = -1.34$ bps, $t = -1.19$) provides the sharpest rebuttal to this concern: Treasury-based and non-Treasury spreads were on the same trend before the SLR event and diverged only when the relief took effect.

Second, as detailed in Section 1.5.2, the pre-period for the entry event coincides with the COVID-19 stress episode. I address this by treating the exit event as the primary identification and the clean DiD (Section 1.5.1) as the whole-sample complement.

Third, the CIP-EUR series contains a single extreme observation at 432.4 basis points (absolute value) in March 2020, driven by the near-breakdown of the cross-currency basis market. This observation is winsorized at the within-series 99th-percentile bound of 432.41 bps (see Table 10). My exit-event estimates use none of this extreme observation in their sample window.

Fourth, the DIRECT specification includes rolling 7-, 14-, and 30-calendar-day total Treasury issuance (in billions, lagged one day for point-in-time discipline) constructed from auction records by maturity bucket. These controls are included to isolate the balance-sheet channel from mechanical Treasury supply effects. Their estimated coefficients are economically small and statistically indistinguishable from zero across all exit specifications (e.g., $W = 60$ DIRECT: $\hat{\delta}_7 = -0.001$ bps per billion, SE = 0.002; $\hat{\delta}_{30} = -0.003$ bps per billion, SE = 0.004), and inclusion of these controls does not meaningfully change the key Treasury interaction (see Table 9). The absence of a supply effect on the Treasury differential is consistent with the balance-sheet channel: the SLR relief operates through intermediary leverage capacity, not through the supply of Treasury collateral.

Fifth, Cochran et al. [7] examine dealer Treasury holdings from regulatory reports and find no noticeable change in aggregate dealer Treasury inventory attributable to the SLR exclusion, which they interpret as evidence against a supply-constraint channel. My results are not necessarily inconsistent with theirs: aggregate inventory is a different object from spread levels, and a marginal reduction in dealer balance-sheet costs can tighten arbitrage spreads without requiring a discrete change in reported inventory positions. Moreover, my identification relies on the differential response of Treasury versus non-Treasury spreads at two specific event dates, which is a sharper test than the aggregate inventory comparison. I acknowledge this as an empirical tension worth investigating with more granular Y-9C or FR 2004 position data.

A sixth technical concern is that five Newey-West lags may be insufficient given AR(1) coefficients exceeding 0.93, at which lag-5 autocorrelation is approximately $0.93^5 \approx 0.70$. Applying the rule of thumb of $T^{1/4}$ lags for a sample of $T \approx 121$ trading days suggests $\approx$ 3–4 lags for the event-window regressions, consistent with the choice of five. The Driscoll-Kraay results (which strengthen the exit Treasury interaction to $t_{\text{DK}} = 5.62$) provide reassurance that the primary inference is not sensitive to the autocorrelation correction, though a formal sensitivity analysis with 20 or more lags would further address this concern for the long time-series panels used in the DiD.

Conclusion

This paper investigates whether regulatory leverage constraints are the operative friction behind Treasury market arbitrage dislocations. Using the Federal Reserve's 2020–2021 SLR exclusion as a time-dated quasi-experiment, I find that the expiry of regulatory relief caused Treasury-based arbitrage spreads to widen by 3.84 basis points more than non-Treasury spreads in the subsequent 60 trading days ($t = 4.32$, $p < 0.01$), with the estimate stable to the one-hundredth of a basis point across control structures and robust to cross-sectional dependence (Driscoll-Kraay $t = 5.62$). A full-period difference-in-differences recovers -10.76 basis points of differential Treasury compression during the relief window relative to the 2019 pre-COVID baseline ($t = -9.37$). Non-Treasury series—CIP deviations for eight G-10 currencies and equity spot-futures for three major indices—show no comparable break at the SLR event dates, passing the falsification tests that are the design's primary identification safeguard.

These results advance the work of Siriwardane et al. [1] in a specific direction. The authors document that the average pairwise correlation among 32 arbitrage spreads is only 22 percent, in-

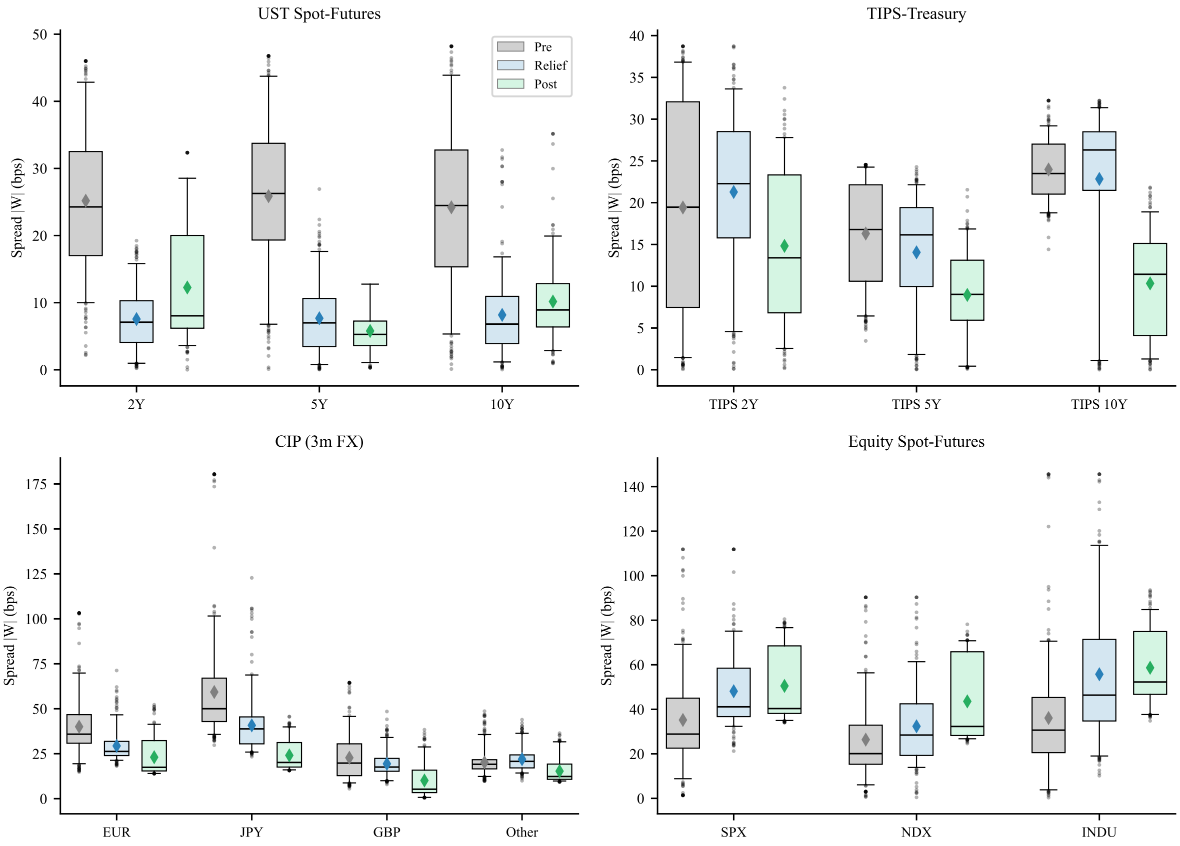

Figure 6: Distribution of absolute arbitrage spreads $|W|$ by regime. Each box shows the interquartile range with median line; whiskers extend to 1.5×IQR. Treasury-based series (UST spot-futures and TIPS-Treasury) are shown in the left panels; non-Treasury series (CIP deviations and equity spot-futures) in the right panels. Regimes: Pre = January 2019 – March 2020; Relief = April 2020 – March 2021; Post = April 2021 – December 2021 (February–March 2020 excluded from DiD regressions but included here for visual completeness). The sharp compression of Treasury-based spreads during the relief period and partial re-widening at expiry are visible relative to the stable non-Treasury distributions.

consistent with representative-intermediary models, and attribute this to segmented funding and segmented balance sheets. Whereas Siriwardane et al. [1] document segmentation and attribute it to balance-sheet constraints using cross-sectional and time-series variation over 2010–2020, this study uses a single policy event to directly test whether a regulatory relaxation of those constraints produces the predicted differential response across strategy types. My results indicate that primary dealer balance sheets are indeed marginal for Treasury spot-futures arbitrage—the most balance-sheet-intensive of the Treasury strategies I study—and that the regulatory effect is detectable in market prices within a 20-trading-day window. The UST spot-futures series compressed by approximately 70 percent during the relief period (pre-period mean 25.6 bps to 7.8 bps during relief), and the cross-strategy divergence at expiry is economically meaningful relative to the 26–33 percent shadow cost of primary dealer leverage constraints estimated by Bräuning and Stein [2].

The results also speak to an active policy debate. In 2025, the Federal Reserve proposed reforms to the enhanced SLR buffer for G-SIBs that would free substantial capital at G-SIB depository subsidiaries. The 2020–2021 episode provides the only available quasi-experimental benchmark for the price-impact magnitude of such a reform. My estimates suggest that a comparable structural change in leverage capacity could compress Treasury-based arbitrage spreads by several basis points on a sustained basis, reducing the intermediation cost embedded in Treasury market prices.

Three extensions remain for future work. First, a bank-level mechanism test using quarterly FR Y-9C data would directly link the aggregate spread compression to variation in individual banks' SLR bind, providing the cross-sectional identification that the current design cannot. Second, the partial reversion evidence—the point estimate from the full-period DiD implies approximately 16 percent of the relief-period compression reversed in the nine-month post-expiry window, though this figure is statistically imprecise ($p = 0.11$; the 95% CI spans essentially zero to -3.86 bps)—suggests that the structural impact of the exclusion, combined with concurrent Fed balance sheet expansion and the launch of the Standing Repo Facility, may have durably improved Treasury market intermediation in ways that outlasted the formal regulation. Third, a local projections specification would allow the impulse response of each spread to be estimated flexibly over post-event horizons, which would be informative about the speed of arbitrage capital reallocation following the regulatory change.

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Table 7: Summary Statistics for Arbitrage Spreads

Strategy	Series	N	Mean $\overline{W}$	Median W	Mean $\lvert W \rvert$	$Q_{0.05}(W)$	$Q_{0.95}(W)$
Panel A: Treasury-based arbitrages (TreasuryBased = 1)							
TIPS–Treasury	2y, pre	312	16.6	19.5	19.4	−9.7	36.8
	2y, relief	250	16.8	22.3	21.4	−16.7	33.7
	2y, post	190	11.7	13.4	14.8	−11.1	27.8
	5y, pre	312	16.3	16.8	16.3	6.4	24.3
	5y, relief	250	13.7	16.2	14.1	−2.1	22.3
	5y, post	190	8.7	9.0	9.0	−0.4	16.9
	10y, pre	312	24.1	23.5	24.1	18.8	29.2
	10y, relief	250	22.7	26.3	22.9	0.1	31.4
	10y, post	190	10.2	11.4	10.4	−0.6	19.2
UST spot–fut.	2y, pre	314	−25.4	−24.3	25.6	−43.0	−9.6
	2y, relief	251	−5.8	−6.0	7.6	−16.1	9.4
	2y, post	192	10.0	8.0	12.6	−3.2	27.4
	5y, pre	314	−26.3	−26.3	26.4	−43.8	−6.6
	5y, relief	251	−6.3	−6.5	7.7	−17.8	5.9
	5y, post	192	5.5	5.3	5.8	−0.4	12.8
	10y, pre	314	−20.8	−23.4	24.7	−39.1	6.9
	10y, relief	251	−5.4	−6.1	8.2	−16.0	7.7
	10y, post	192	10.1	8.9	10.2	2.8	20.1
Panel B: Non-Treasury arbitrages (TreasuryBased = 0, means across tenors/currencies)							
CIP 3m (8 curr.)	pre	326 avg.	—	—	31.1	—	—
	relief	261 avg.	—	—	25.9	—	—
	post	197 avg.	—	—	17.2	—	—
Equity SF (3 idx.)	pre	310 avg.	—	—	32.8	—	—
	relief	248 avg.	—	—	45.8	—	—
	post	189 avg.	—	—	50.9	—	—

Notes. Spread statistics are in basis points. $\overline{W}$ denotes the mean signed spread; $\lvert W \rvert$ denotes the mean absolute spread. Regime boundaries: pre = January 1, 2019 to March 31, 2020; relief = April 1, 2020 to March 31, 2021; post = April 1 to December 31, 2021. CIP and equity rows report cross-series averages within each regime. All series are winsorized at the within-series first and 99th percentiles prior to analysis; see Table 10. *Note*: the 'pre' regime includes February–March 2020 (the acute COVID crash), which are excluded from the DiD regressions in Table 8 Panel B. Pre-period means for UST spot-futures are therefore modestly inflated by COVID-period observations relative to the 2019-only baseline used in the DiD.

Table 8: SLR Exclusion Event Study: Pooled Jump Regressions

	Entry event (2020-04-01)†				Exit event (2021-03-31)			
	$W = 20$		$W = 60$		$W = 20$		$W = 60$	
	Total	Direct	Total	Direct	Total	Direct	Total	Direct
Panel A: Pooled regressions								
Post	-0.84	$+1.45$	-7.08^{**}	-2.76	-2.62^{***}	-2.28^{***}	-3.09^{***}	-2.97^{***}
	(4.77)	(7.03)	(3.05)	(5.54)	(0.77)	(0.68)	(0.83)	(0.85)
Post × TreasuryBased	-2.16	-2.23	-4.54	-4.52	$+3.91^{***}$	$+3.90^{***}$	$+3.84^{***}$	$+3.84^{***}$
	(6.13)	(6.14)	(3.00)	(3.00)	(0.92)	(0.92)	(0.89)	(0.88)
N	697	697	2,057	2,057	697	697	2,057	2,057
R^2	0.61	0.61	0.55	0.55	0.96	0.96	0.89	0.89
Panel B: Robust DiD (2019 pre-COVID baseline, excl. Feb–Mar 2020)								
	Relief period				Post-relief period			
	Total		Direct		Total		Direct	
Non-Treasury baseline	-1.34		-11.03^{***}		-6.92^{***}		-14.87^{***}	
	(1.13)		(2.97)		(1.01)		(2.80)	
× TreasuryBased	-10.76^{***}		-10.77^{***}		-9.03^{***}		-9.04^{***}	
	(1.15)		(1.14)		(1.16)		(1.16)	
N	12,326				12,326			
R^2	0.516				0.516			

Notes. Dependent variable is $|W_{i,t}|$, the absolute spread in basis points. Post = 1 for event-time ≥ 0 within the stated $\pm W$ trading-day window. TreasuryBased = 1 for UST spot-futures and TIPS-Treasury series. All specifications include series fixed effects. Total controls: VIX, HY OAS, Baa-10y spread. Direct adds SOFR, TGCR-SOFR, and rolling 7/14/30-day Treasury issuance. HAC standard errors (Newey-West, 5 lags) in parentheses. $^{***}\,p < 0.01$, $^{**}\,p < 0.05$, $^{*}\,p < 0.10$. † The entry pre-period (Jan–Mar 2020) spans the COVID-19 crash peak; entry estimates are contaminated by contemporaneous Fed interventions. See Section 1.5.2 and Panel B (Robust DiD) for COVID-robust estimates. The identical $R^2 = 0.516$ across Total and Direct in Panel B reflects that the additional Direct controls (SOFR, TGCR-SOFR, issuance) add negligible explanatory power once series fixed effects and the broad risk controls absorb common daily variation; the controls affect the partition of coefficients between $\hat{\mu}_1$ and the regime dummies, not the overall fit. The large Direct non-Treasury baseline during relief (-11.03 bps) is discussed in Section 1.5.1; the Treasury interaction $\hat{\mu}_2$ is stable across specifications (-10.76 vs. -10.77 bps), confirming the Treasury differential is orthogonal to the secured funding channel.

Table 9: Robustness: Total vs. Direct Specification Comparison

Event	Window	Coefficient	N	$\hat{\beta}_{\text{Tot}}$	SE_{Tot}	t_{Tot}	Stars	$\hat{\beta}_{\text{Dir}}$	SE_{Dir}	t_{Dir}	$\Delta\hat{\beta}$	Δt
Entry	$W = 20$	Post	697	-0.84	4.77	-0.18	—	$+1.45$	7.03	$+0.21$	$+2.29$	$+0.38$
Entry	$W = 20$	Post × Treas.	697	-2.16	6.13	-0.35	—	-2.23	6.14	-0.36	-0.06	-0.01
Entry	$W = 60$	Post	2,057	-7.08	3.05	-2.32	**	-2.76	5.54	-0.50	$+4.32$	$+1.82$
Entry	$W = 60$	Post × Treas.	2,057	-4.54	3.00	-1.51	—	-4.52	3.00	-1.51	$+0.02$	$+0.00$
Exit	$W = 20$	Post	697	-2.62	0.77	-3.40	***	-2.28	0.68	-3.33	$+0.34$	$+0.06$
Exit	$W = 20$	Post × Treas.	697	$+3.91$	0.92	$+4.26$	***	$+3.90$	0.92	$+4.24$	-0.01	-0.02
Exit	$W = 60$	Post	2,057	-3.09	0.83	-3.72	***	-2.97	0.85	-3.48	$+0.12$	$+0.24$
Exit	$W = 60$	Post × Treas.	2,057	$+3.84$	0.89	$+4.32$	***	$+3.84$	0.88	$+4.34$	0.00	$+0.02$

Notes. Each row reports both Total and Direct estimates of the stated coefficient from regression (1). "Post × Treas." abbreviates $\text{Post}_t \times \text{TreasuryBased}_i$. Stars correspond to the Total specification; Direct significance is identical for all exit rows. HAC SE (Newey-West, 5 lags). $^{***}\,p < 0.01$, $^{**}\,p < 0.05$.

Table 10: Winsorization Log: Per-Series Clipping Counts

Series	Variable	p_1 bound	p_{99} bound	$n_{\text{clipped, lo}}$	$n_{\text{clipped, hi}}$	% clipped
CIP-AUD	$\|W\|$	0.25	44.60	40	40	2.04
CIP-CAD	$\|W\|$	0.24	45.14	40	40	2.04
CIP-CHF	$\|W\|$	0.91	103.72	40	40	2.05
CIP-EUR	$\|W\|$	1.58	432.41	40	40	2.04
CIP-GBP	$\|W\|$	0.81	55.50	40	40	2.04
CIP-JPY	$\|W\|$	12.67	110.43	40	40	2.05
CIP-NZD	$\|W\|$	0.52	41.22	40	40	2.04
CIP-SEK	$\|W\|$	1.11	92.02	40	40	2.04
Eq. INDU	$\|W\|$	3.07	126.51	38	38	2.03
Eq. NDX	$\|W\|$	1.88	109.21	38	38	2.04
Eq. SPX	$\|W\|$	3.53	111.44	38	38	2.04
TIPS 10y	$\|W\|$	1.66	38.38	38	38	2.03
TIPS 2y	$\|W\|$	0.45	42.05	38	38	2.04
TIPS 5y	$\|W\|$	0.77	38.45	38	38	2.03
UST SF 10y	$\|W\|$	0.51	227.33	50	50	2.01
UST SF 2y	$\|W\|$	0.42	178.09	50	50	2.01
UST SF 5y	$\|W\|$	0.27	179.19	50	50	2.01

Notes. Each series is winsorized within-series at the first and 99th percentiles prior to all estimation. p_1 and p_{99} are the clipping bounds in basis points. "% clipped" reports the fraction of observations replaced at either tail. Total clipped observations in the $|W|$ dependent variable across the full stacked panel of 68,620 series-day pairs: approximately 1,396 (2.04 percent). The pipeline also winsorizes the signed spread W separately; counting both variables gives approximately 2,792 total clips, but the relevant figure for the estimation sample is 1,396.